

CH₄ / C₂H₄ Splitter Shortcut FUG Validation

Calculation Record

Project	Column	Pressure	Service
Splitter design review	CH ₄ / C ₂ H ₄ splitter	7.0 bar	Methane / ethene separation

Introduction

This calculation record documents the shortcut Fenske-Underwood-Gilliland validation basis used for the project review package. The accompanying executive memo summarizes the decision-facing results and limitations; this record preserves the traceable calculations and conventions behind those results.

Precision convention. Source values are shown at the precision supplied. Derived intermediate quantities are carried without premature rounding and are displayed here at extended precision so each step can be manually checked. Final reported values follow the task requirement: theta, R_{min}, R, mole fractions, D/F and B/F to four decimal places; N_{min} and the Gilliland/Eduljee stage estimate to two decimal places.

1. Case basis

Item	Basis used
Column configuration	Total condenser; kettle reboiler
Operating pressure	7.0 bar
Feed rate	24,000 kg/h
Feed mass fractions	CH ₄ = 0.62; C ₂ H ₄ = 0.38
Feed thermal condition	Subcooled; q = 1.2
Relative volatility	alpha_top = 2.500 (distillate-end); alpha_bottom = 1.369 (bottoms-end); effective alpha = 1.850000
Keys	Light key = CH ₄ ; heavy key = C ₂ H ₄
Overhead GC ratio	CH ₄ :C ₂ H ₄ = 18.0:1.00; normalize directly as distillate composition
Bottoms composition	x _{B,CH₄} = 0.0100; x _{B,C₂H₄} = 0.9900

Effective relative volatility. The volatility varies across the column, so the shortcut method uses a single representative value taken as the geometric mean of the distillate-end and bottoms-end values, per the standard Fenske averaging convention:

$\alpha_{\text{eff}} = \sqrt{\alpha_{\text{top}} \times \alpha_{\text{bottom}}} = \sqrt{2.500 \times 1.369} = \sqrt{3.4225} = 1.850000$

This effective value ($\alpha_{\text{eff}} = 1.850000$) is used as the constant relative volatility in the Fenske and Underwood steps below.

2. Component reference data

Component	Molar mass (g/mol)	Case role	Source
CH ₄ (methane)	16.043	Light key	CH ₄ _C ₂ H ₄ _Component_Reference.docx
C ₂ H ₄ (ethene)	28.054	Heavy key	CH ₄ _C ₂ H ₄ _Component_Reference.docx

3. Mass-to-mole conversion and feed mole fractions

Component mass flows from the 24,000 kg/h feed:

$m_{\text{CH}_4} = 24,000 \times 0.62 = 14,880 \text{ kg/h}$

$m_{\text{C}_2\text{H}_4} = 24,000 \times 0.38 = 9,120 \text{ kg/h}$

Convert each component to molar flow using the supplied molar masses:

$n_{\text{CH}_4} = 14,880 / 16.043 = 927.507324066571 \text{ kmol/h}$

$n_{\text{C}_2\text{H}_4} = 9,120 / 28.054 = 325.087331574820 \text{ kmol/h}$

$F_{\text{molar}} = 927.507324066571 + 325.087331574820 = 1,252.594655641391 \text{ kmol/h}$

$z_{\text{CH}_4} = 927.507324066571 / 1,252.594655641391 = 0.740468849910 \rightarrow \text{report } 0.7405$

$z_{\text{C}_2\text{H}_4} = 325.087331574820 / 1,252.594655641391 = 0.259531150090 \rightarrow \text{report } 0.2595$

4. Distillate composition derived from the GC ratio

The GC is on the vapor line above the total condenser. Per the case-data convention, normalize the reported 18.0:1.00 vapor peak ratio directly and use the normalized values as the distillate composition.

$$x_{D,CH_4} = 18.0 / (18.0 + 1.00) = 0.947368421053 \rightarrow \text{report } 0.9474$$
$$x_{D,C_2H_4} = 1.00 / (18.0 + 1.00) = 0.052631578947 \rightarrow \text{report } 0.0526$$

5. Overall and component molar balance

Let $d = D/F$ and $b = B/F$ on a molar basis. The methane balance is:

$$z_{CH_4} = d \cdot x_{D,CH_4} + b \cdot x_{B,CH_4}, \text{ with } b = 1 - d$$
$$d = (z_{CH_4} - x_{B,CH_4}) / (x_{D,CH_4} - x_{B,CH_4})$$
$$= (0.740468849910 - 0.0100) / (0.947368421053 - 0.0100)$$
$$= 0.779276145328 \rightarrow \text{report } D/F = 0.7793$$
$$b = 1 - 0.779276145328 = 0.220723854672 \rightarrow \text{report } B/F = 0.2207$$

Component-balance checks using the full-precision d and b values:

$$CH_4 \text{ check} = d \cdot x_{D,CH_4} + b \cdot x_{B,CH_4}$$
$$= 0.779276145328 \cdot 0.947368421053 + 0.220723854672 \cdot 0.0100$$
$$= 0.740468849910 \rightarrow \text{report } 0.7405 = z_{CH_4}$$

$$C_2H_4 \text{ check} = d \cdot x_{D,C_2H_4} + b \cdot x_{B,C_2H_4}$$
$$= 0.779276145328 \cdot 0.052631578947 + 0.220723854672 \cdot 0.9900$$
$$= 0.259531150090 \rightarrow \text{report } 0.2595 = z_{C_2H_4}$$

$$\text{Overall closure} = D/F + B/F = 0.779276145328 + 0.220723854672 = 1.000000000000$$

6. Fenske calculation of N_{min} (including reboiler)

Use the light-key/heavy-key form with the effective relative volatility $\alpha_{eff} = 1.850000$ from Section 1. The kettle reboiler counts as one equilibrium stage; the total condenser does not.

$$N_{min} = \ln[(x_{D,LK}/x_{B,LK}) \cdot (x_{B,HK}/x_{D,HK})] / \ln(\alpha_{eff})$$
$$x_{D,LK}/x_{B,LK} = 0.947368421053 / 0.0100 = 94.736842105263$$
$$x_{B,HK}/x_{D,HK} = 0.9900 / 0.052631578947 = 18.810000000000$$
$$\text{Product ratio} = 94.736842105263 \times 18.810000000000 = 1782.000000000000$$
$$\ln(1782) = 7.485491608031$$
$$\ln(\alpha_{eff}) = \ln(1.850000) = 0.615185639090$$
$$N_{min} = 7.485491608031 / 0.615185639090 = 12.167858175462 \rightarrow \text{report } 12.17 \text{ stages}$$

7. Underwood calculation of Rmin

For $q = 1.2$, $\alpha_{HK} = 1.000$ and $\alpha_{LK} = \alpha_{eff} = 1.850000$ (Section 1). Use:

$$\sum[\alpha_i z_i / (\alpha_i - \theta)] = 1 - q = -0.2$$

Substitute the full-precision feed mole fractions:

$$1.85 * 0.740468849910 / (1.85 - \theta) + 1.000 * 0.259531150090 / (1.000 - \theta) = -0.2$$

Clearing the denominators gives the equivalent quadratic:

$$\theta^2 - 10.996992612119 * \theta + 11.100000000000 = 0$$

The two mathematical roots are approximately 1.124315053785 and 9.872677558334. The required root lies between α_{HK} and α_{LK} , so:

$$\theta = 1.124315053785 \rightarrow \text{report } 1.1243$$

Back-substitution into the feed Underwood equation:

$$\text{LK term} = 1.85 * 0.740468849910 / (1.85 - 1.124315053785) = 1.887688837255$$

$$\text{HK term} = 1.00 * 0.259531150090 / (1.00 - 1.124315053785) = -2.087688837255$$

$$\text{Sum} = 1.887688837255 - 2.087688837255 = -0.200000000000 = 1 - q$$

Now compute the minimum reflux ratio:

$$R_{min} + 1 = \sum[\alpha_i x_{D,i} / (\alpha_i - \theta)]$$

$$\text{LK term} = 1.85 * 0.947368421053 / (1.85 - 1.124315053785) = 2.415141154696$$

$$\text{HK term} = 1.00 * 0.052631578947 / (1.00 - 1.124315053785) = -0.423372530879$$

$$R_{min} + 1 = 2.415141154696 - 0.423372530879 = 1.991768623817$$

$$R_{min} = 1.991768623817 - 1 = 0.991768623817 \rightarrow \text{report } 0.9918$$

8. Gilliland calculation using the Eduljee explicit form

Use the specified operating reflux ratio $R = 1.25 \times R_{min}$ and the Eduljee explicit correlation from FUG_inputs.pdf.

$$R = 1.25 \times 0.991768623817 = 1.239710779771 \rightarrow \text{report } 1.2397$$

$$X = (R - R_{min}) / (R + 1)$$

$$= (1.239710779771 - 0.991768623817) / (1.239710779771 + 1)$$

$$= 0.110702755996$$

$$X^{0.5668} = 0.287229471952$$

$$Y = 0.75 * (1 - X^{0.5668}) = 0.534577896036$$

$$N = (N_{min} + Y) / (1 - Y)$$

$$= (12.167858175462 + 0.534577896036) / (1 - 0.534577896036)$$

$$= 27.292292229592 \rightarrow \text{report } 27.29 \text{ theoretical stages}$$

9. Verification block - Underwood back-substitution

The verification below uses the full-precision theta and Rmin values from Sections 7 and 8, not the rounded reporting values.

Check	Full-precision result	Expected / closure	Reported
Feed Underwood equation	-0.200000000000	$1 - q = -0.200000000000$	-0.2000
Distillate reflux equation	1.991768623817	$R_{min} + 1 = 1.991768623817$	1.9918
Rmin from reflux equation	0.991768623817	Same full-precision Rmin	0.9918
CH ₄ component balance	0.740468849910	Feed $z_{CH_4} = 0.740468849910$	0.7405
C ₂ H ₄ component balance	0.259531150090	Feed $z_{C_2H_4} = 0.259531150090$	0.2595
Overall balance	1.000000000000	$D/F + B/F = 1$	1.0000

10. Final results summary table

Final reported values below follow the task-specified reporting precision.

Quantity	Final reported value	Units
Feed mole fraction z_{CH_4}	0.7405	dimensionless
Feed mole fraction $z_{C_2H_4}$	0.2595	dimensionless
Distillate mole fraction x_{D,CH_4}	0.9474	dimensionless
Distillate mole fraction x_{D,C_2H_4}	0.0526	dimensionless
Bottoms mole fraction x_{B,CH_4}	0.0100	dimensionless
Bottoms mole fraction x_{B,C_2H_4}	0.9900	dimensionless
Effective relative volatility (α_{eff})	1.8500	dimensionless
Underwood root (theta)	1.1243	dimensionless
Minimum reflux ratio (R_{min})	0.9918	dimensionless
Operating reflux ratio ($R = 1.25 \times R_{min}$)	1.2397	dimensionless
Minimum number of stages (N_{min} , including reboiler)	12.17	stages

Gilliland/Eduljee theoretical stage estimate	27.29	stages
Distillate-to-feed ratio (D/F, molar basis)	0.7793	dimensionless
Bottoms-to-feed ratio (B/F, molar basis)	0.2207	dimensionless

Interpretation note. The Gilliland/Eduljee result is a continuous theoretical equilibrium-stage estimate at the specified operating reflux. It is not an installed tray count and should not be used by itself to judge final column adequacy.